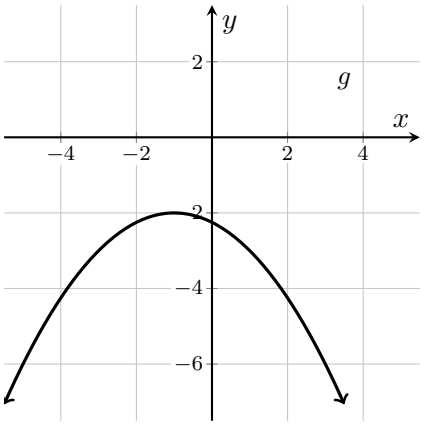
Name _____ Period _____ Date _____

1-2 Lesson Quiz: Transformations of Functions

short form

1. Select **all** the transformations of $f(x) = x^2$ that combine to result in the graph of function g at the right.

- ☐ **A.** translation of 1 unit left
- ☐ **B.** translation of 1 unit right
- ☐ **C.** translation of 2 units down
- ☐ **D.** horizontal stretch by a factor of 2
- ☐ **E.** vertical stretch by a factor of 2
- ☐ **F.** reflection across the x-axis



2. The graph represents f. Graph $g(x) = 2f(x-1)$ on the same grid.

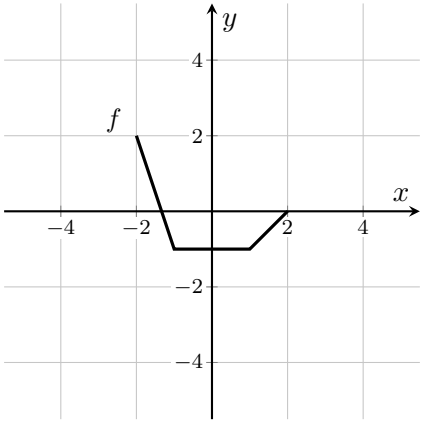
Show your thinking. For one point of f, write where it lands on g:

(_____ , _____) on f $\longrightarrow$ (_____ , _____) on g

Which transformation did you do first, the stretch or the shift? Why?

_____

_____

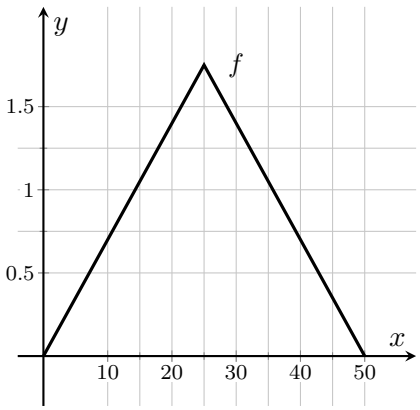


3. The graph of the function f is shown. The domain of f is $[0, 50]$. What is the range of $g(x) = 4f(x)$?

- Ⓐ $[0, 50]$
- Ⓑ $[0, 70]$
- Ⓒ $[0, 1.75]$
- Ⓓ $[0, 7]$

Explain in one sentence why the domain of g is still $[0, 50]$:

_____



Scoring: Question 2 and Question 3 are 10 points each. Question 1 is a bonus of up to 5 points (all four correct boxes, none wrong). Total out of 20.

1-2 Lesson Quiz, short form: answer key

---

1. **A, C, D, F.** Vertex $(-1, -2)$: left 1, down 2. Opens down: reflection across the x-axis. Wider than x^2 (passes through $(1, -3)$, so $a = -\frac{1}{4}$): horizontal stretch by 2. Savvas quiz item 1, DOK 2, standard HSF.BF.B.3. **Bonus 5:** all four boxes and no extra box; 3 for three correct and none wrong; 0 otherwise.

2. g **joins** $(-1, 4)$**,** $(0, -2)$**,** $(2, -2)$**,** $(3, 0)$**.** Each point (x, y) of f goes to $(x + 1,\ 2y)$. Either order of stretch and shift gives the same graph here because the stretch is vertical and the shift is horizontal; accept either order with a reason. Savvas quiz item 3, DOK 1, HSF.BF.B.3. **10 points:** 6 for the four correct points, 2 for the mapped point, 2 for the order explanation.

3. **D,** $[0, 7]$**.** The maximum of f is 1.75 and $4(1.75) = 7$. The domain is unchanged because $4f(x)$ multiplies outputs, not inputs. Savvas quiz item 5, DOK 2, HSF.IF.B.5 with HSF.BF.B.3. **10 points:** 7 for D, 3 for the domain sentence naming outputs versus inputs.

Items left off this form (available for a retake or Form B): Savvas item 2, equation of $2x^2 + 3x - 1$ reflected across the y-axis (answer C, $2x^2 - 3x - 1$); Savvas item 4, the leaking-bucket translation (answer D, $|x - 3|$).

IXL replacement: SmartScore 70 or above on NNY, RSN and KQL replaces this quiz with 20/20.